

College of Engineering

Coursework Submission Sheet

Coursework Title: C1-3 Coursework - Group Project Report on Thrust Vector control for Model Rockets	Student number: 2101048, 2502923, 2380424, 2456476, 2428966
Coursework number (i.e., CW1 CW2) C1-1	Name: Arsenios Konstantinou
Module code: EG-M37	Shreyas Ramesh Shetty
Module title: Additive Manufacturing	Praveen Kumar Mohandas
Submission deadline: 4th April 2025	Varghese Mathew
Lecturer: Prof. Nick Lavery	Akshay Ravindra Taware
	Degree course: MSc Mechanical Engineering

By submitting this coursework, I certify that this is all my own work.

Submission date

Student signature if this is a non-electronic submission.....

SPLD Students

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Agreed group effort.

Member	1	2	3	4	5
Student ID	2101048	2502923	2380424	2456476	2428966
Effort (%)	100	100	100	100	100
Signature	AK	SRS	PKM	VM	ART

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EXECUTIVE SUMMARY

This project consists of the development of a compact thrust vector control (TVC) system designed for model rockets. The TVC system provides directional control over the trajectory of the rocket by mechanically redirecting thrust using two servo motors, improving the overall maneuverability compared to traditional fin-based stabilization. The integration of advanced design for AM sets this project apart as it enables the creation of complex designs, such as spiral reinforcement structures, which would be difficult to produce using traditional manufacturing methods. The use of Ti-6Al-4V, further enhances the mechanical properties of the system while maintaining low mass, which is an essential factor in aerospace applications. In addition, the heat resistance of the titanium alloy enables longer operations. The mechanical integrity of each component was validated through SolidWorks simulations under a peak thrust of 12.1 N [6], representative of the Estes B6-2 motor. Key successes of the project include reduced supports through optimized part orientation, minimal post processing and a significant weight reduction via lattice and hollow features. These achievements indicate the possible scalability of our design as a cost-effective aerospace component as well as the potential of metal additive manufacturing to revolutionize small-scale aerospace control systems.

PROJECT: THRUST VECTOR CONTROL (TVC) FOR MODEL ROCKETS

CHAPTER - 1

1.1. INTRODUCTION:

Developments in technologies related to rocketry combined with additive manufacturing methods have revolutionized the aerospace field. Especially, rocket maneuvering and stabilizing technologies have played a significant role in this progress. One of the key advancements is the Thrust Vector control (TVC) system that allows a rocket to control its trajectory by adjusting the direction of its thrust [1]. This project involves the design improvement and manufacturing of TVC components through 3D printing techniques.

1.2. OBJECTIVES OF THE PROJECT:

The primary purpose of this project is to design and manufacture a lightweight, durable, and precisely functional Thrust Vector Control system using Additive Manufacturing (AM) methods. The project also includes functionality objectives such as ensuring precision control, flight stability, and accurate adjustment in thrust control during flight. Another manufacturing goal is aimed at creating cost-effective, reliable, and customized components with minimal material wastage. To make component structures visually appealing our design focuses on creating smooth and streamlined parts while prioritizing functionality.

1.3. STRATEGIES FOR DEVELOPING PROJECT IDEAS:

1.3.1. Problem Identification:

The rocket with a fin mechanism offers limited control over stability and maneuvering capability during flight. Additionally, fins and other parts that do not produce thrust are ineffective in space and the upper atmosphere [1]. Another challenge is that manufacturing components with the traditional method involves high cost, material wastage, as well as requires a series of machining and post-processing. With conventional techniques, it is difficult to design customized parts, which are crucial in weight and cost reduction [2].

1.3.2. Solution Exploration:

Technologies such as Thrust Vector Control can significantly enhance stability and flight precision and eliminate the need for a fin mechanism. By implementing the TVC system, rockets can adjust their trajectory quickly and accurately, reducing the risk of deviation from the directed path. Leveraging additive manufacturing (AM) techniques in the project will reduce problems like material waste, structural inconsistency, and post-processing. In addition, the AM allows the creation of complicated and customized parts that are difficult to produce with traditional manufacturing methods [3].

1.4. DESIGN CONSIDERATIONS:

The designs created for this project are based on the constraints outlined in the design brief, ensuring that all parts fit within the allocated 100mm x 100mm build plate area, as well as within the individual part's volume of 20mm x 30mm x 70mm. To minimize post-processing efforts such as drilling or machining, we designed our parts by considering key design principles of additive manufacturing [4]. For the parts with holes, we kept the diameters below 12mm based on the rule of thumb to avoid unnecessary support structures. Additionally, we accounted for the staircase effect to achieve a smoother surface finish with minimal post-processing. To

further enhance the surface finish and reduce support structure, we added features such as fillets and rounded edges instead of sharp angles in the design. We also incorporated a part interlocking strategy into our designs to ease the assembly process. For instance, the inner and outer gimbal can fit into each other easily and require only two screws to assemble them [5].

During the design phase, we also analyzed the impact of support structures on the final print quality and easy removal of finished parts. Using the Netfabb software, we highlighted the areas requiring the support structures and reduced them with correct orientation. The TVC system is an assembly of five components, which include the motor sleeve, inner gimbal, outer gimbal, inner linkage, and outer linkage. For our project, we are using Titanium Alloy (Ti-6Al-4V), a well-suited material for the TVC system due to its ability to achieve a dense structure. This higher level of density will reduce defects like porosity and residual stress, and help in improving tensile strength, corrosion resistance, and other mechanical properties [5].

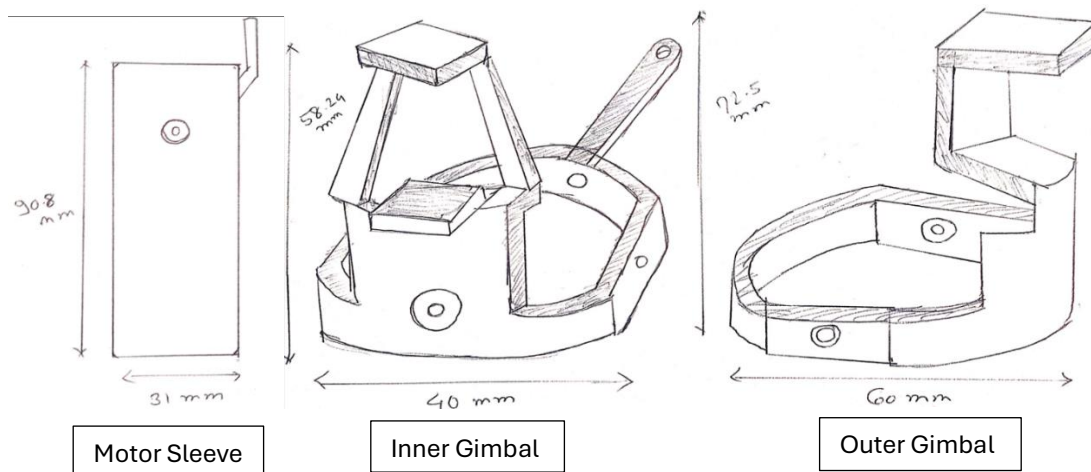
1.5. ADDITIVE MANUFACTURING BENEFITS:

Utilizing additive manufacturing (AM) methods for printing TVC components has several key benefits over conventional manufacturing techniques. One of the main advantages of AM is its ability to create customized and complex parts, which are nearly impossible or require several processes and equipment to manufacture through conventional methods [2,3]. This feature of AM played a significant role in customizing components like the motor sleeve and the gimbal with helical/spiral structures and hollow geometries to reduce weight and make our design lighter and more efficient.

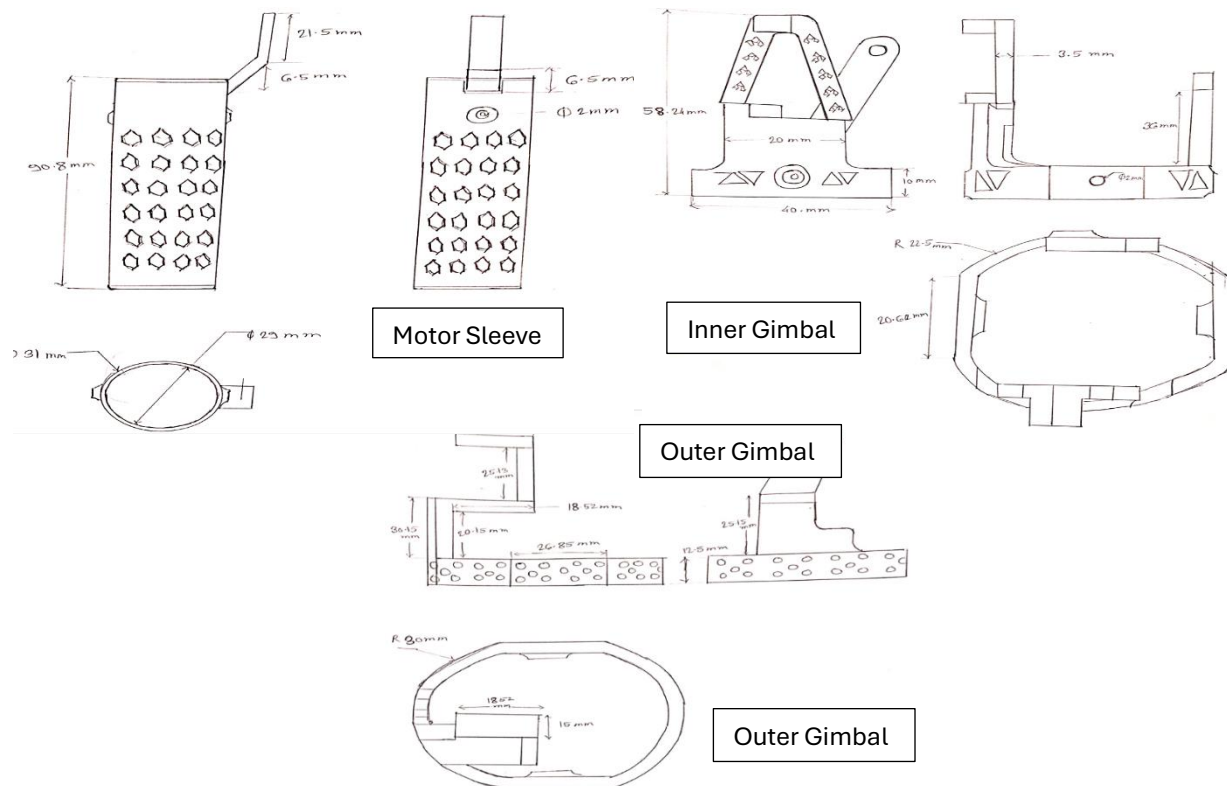
Traditional methods like casting, molding, or the machining process involve substantial wastage of material, which increases the cost. Thus, opting for an AM for our project will reduce the waste of expensive materials like titanium. In conventional manufacturing, changing the size often requires new molds or machining setups, all of which increase lead time and cost. Whereas in AM, scaling a part up or down can be done by modifying the 3D model in CAD software. This is crucial in this project, since different sizes of model rockets are required for different purposes. Methods such as casting and machining often introduce porosity, inaccuracies, and structural weakness in the components, which negatively affect the TVC system. In contrast, AM minimizes these defects by building parts layer by layer with high precision, which not only reduces the risk of fracture and inconsistency but also enhances the mechanical properties of the component [2,3].

1.6. COMMERCIALIZATION POTENTIAL:

When integrated with additive manufacturing, the TVC system has significant potential in many sectors, from aerospace to the defense industry. Especially in aerospace, TVC made with AM will reduce overall project cost and lead time through minimal manufacturing and post-processing. Along with this, customized parts will help in weight reduction and increase payload capacity, which will enhance the fuel efficiency of the rockets or spacecraft. In the field of education and recreational model rocketry, this 3D-printed system will provide a budget-friendly option for students and aerospace enthusiasts. The defense industry could also benefit from this project by enhancing the missile guidance and control system with a lightweight and durable TVC system.

CHAPTER – 2: PRELIMINARY SKETCHES:**2.1. Sketch: I**

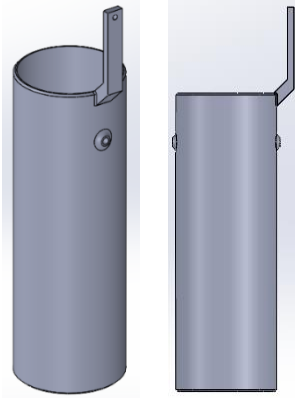
Sketch. I: The first sketch was focused on dimension measurement and analysis of the basic structure of the TVC system to identify the areas for improvement.

2.2. Sketch: II

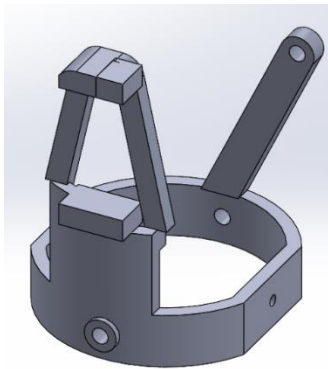
Sketch. II: In this sketch, several refinements and improvements are made to address the limitations in the previous sketch, such as more detailed dimensions, improved aesthetics, and weight reduction through lattice patterns. This redesign also focused on making the body structure more streamlined and smoother with fillets and curved edges to ease the assembly.

3. CONCEPT DESIGN AND PRODUCT DEVELOPMENT CYCLE:

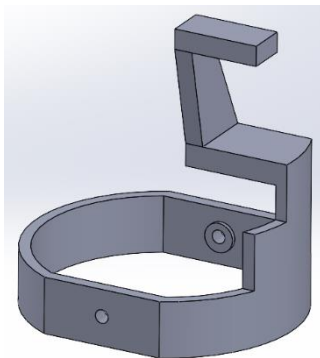
Design iteration 1: Base design



- The motor sleeve was designed with careful consideration of geometrical constraints and dimensions that would have to be finalized before further iterations.
- The initial design was a solid cylindrical structure with necessary holes and cutouts for alignment with the inner gimbal. The assembly involves moving parts, so the radius of the tube to fit the rocket engine motor and the holes to align with the inner gimbal had to be fixed.
- This design was not feasible as it would add a lot of weight to the assembly, considering it would be printed in metal, reducing weight was crucial to enhance efficiency and optimize material usage.



- The inner gimbal is modelled in such a way as to accommodate the motor sleeve inside it while allowing controlled movement within predefined deviation angles.
- The primary goal of this iteration was to define final dimensions and identify potential design challenges that might occur before moving forward with refinements.

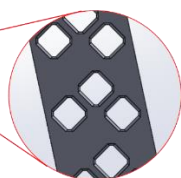
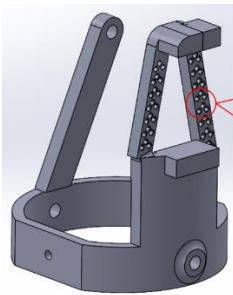


- Similarly, the outer gimbal had to fit within the rocket's body. It must have room to house the servo motors, inner gimbal, and motor sleeve.
- The outer gimbal must be accommodated within the body of the rocket, the radius was assigned by comparing the general radius of the tubes used in model rocket design.
- The geometry was kept simple to focus on achieving proper assembly and functionality.

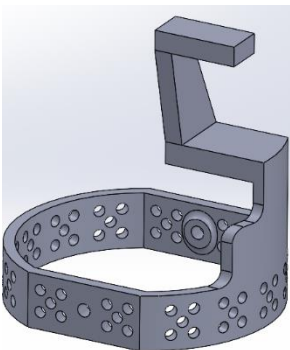
Design iteration 2: Design according to concept sketch



- To address the weight concerns and improve heat dissipation, lattice structures were added in the second iteration.
- These perforations in the sleeve allowed for a significant reduction in weight as well as material usage.
- Additionally, fillets were incorporated at critical points, such as hole edges and corners, to minimize stress concentrations during build. This ensured better structural integrity and reduced the need for excessive support structures, making post-processing easier.

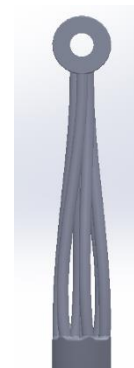
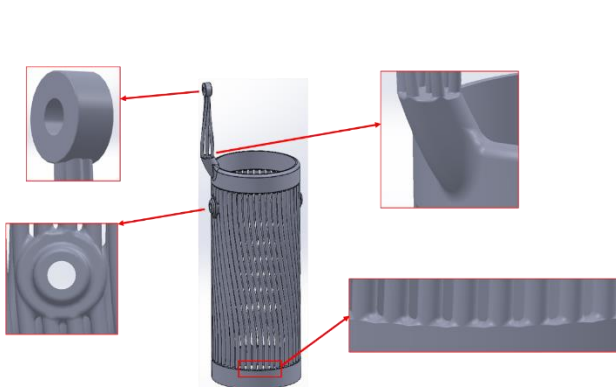


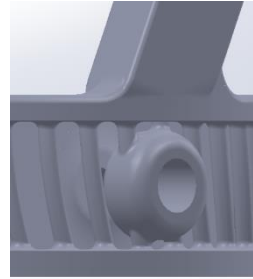
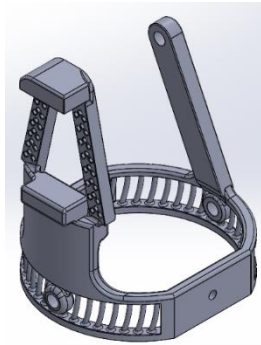
- **Lattice structures** were introduced in areas where high mechanical strength was not a primary requirement.
- These are also added to increase the visual appeal of the part while maintaining the structural strength of the part.
- The overall mass of the part was also reduced, improving the efficiency of the system.



- Circular lattices and attachment holes were integrated into the outer gimbal to facilitate mounting into the rocket body.
- The design needs to be optimized further to facilitate even build and to reduce the number of support structures needed.
- Reducing reliance on support structures will help streamline post-processing and lower material waste.

Design iteration 3: Final design of all parts





- The design was further refined by replacing the lattice structure with a spiral connection/reinforcement pattern.
- This modification provided a better balance between structural strength and weight reduction while also improving build quality.
- The addition of spiral connections helped distribute thermal loads more evenly across the structure, reducing localized stress accumulation that could lead to part failure.
- Another improvement in this iteration was the additional placement of fillets and smooth transitions between connected parts. The smoother geometry also improved printability by minimizing the requirement for extensive support structures, thereby simplifying post-processing.
- Not many changes were made in the outer gimbal as it had to be connected with the rocket body tube. It requires holes as lattice structures.

PRELIMINARY DESIGN REVIEW:

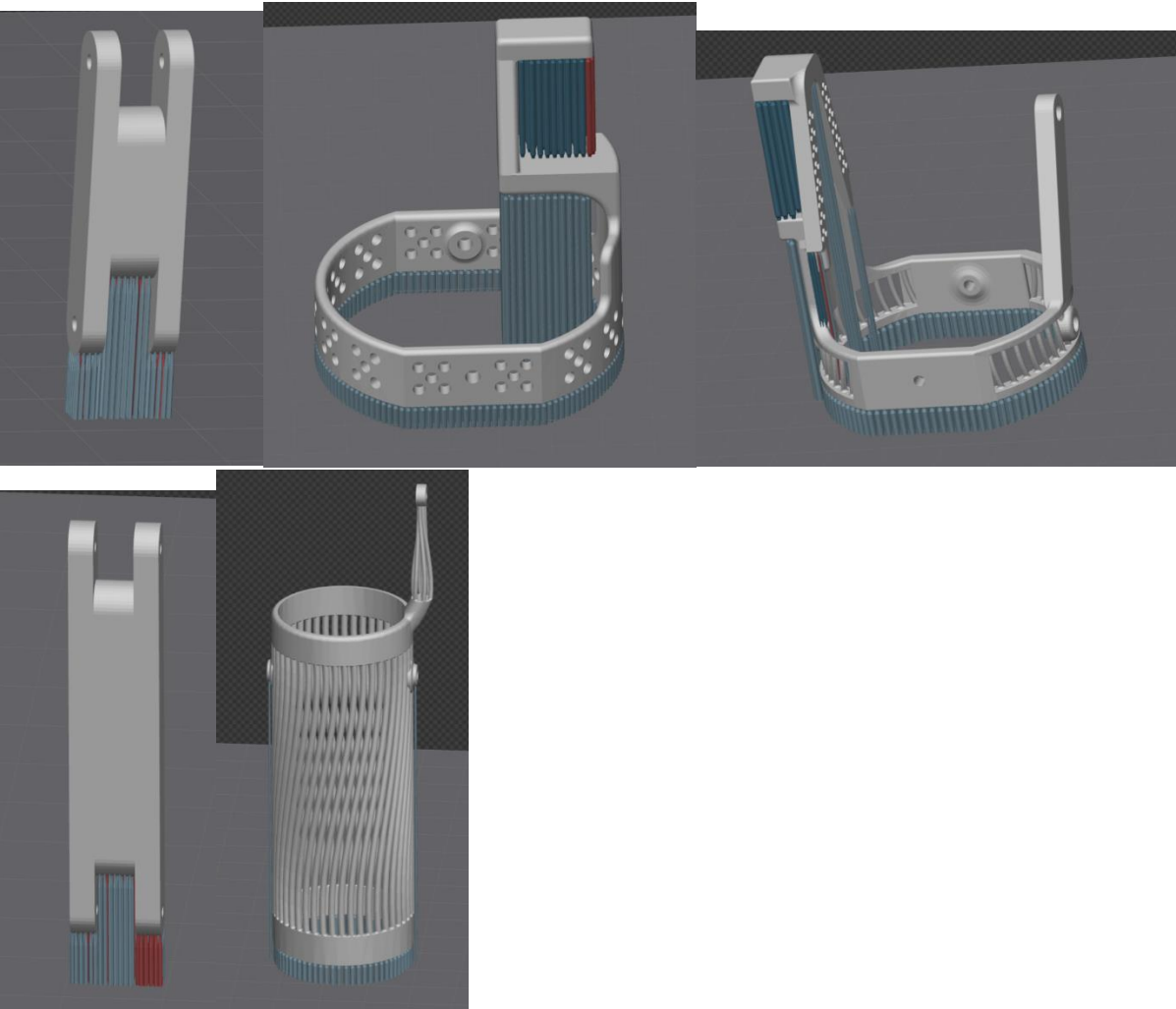
Functional and performance requirements:

- Maximum thrust capacity that the nozzle can produce: 12.1N approximately
- Gimbal angle that can tilt: +15 degree or -15degree
- Operating temperature: 600°C
- Overall weight: 104.5g

How the design is considered:

It is a gimbal-based nozzle actuation that consists of motor sleeve, outer gimbal, inner gimbal, inner linkage arm, outer gimbal ring. The motor sleeve is enclosed within the inner gimbal which is connected to servo motor through linkage which helps to change the direction of thrust in a particular direction and the inner gimbal that is surrounded by outer gimbal allows to rotate in opposite axis. So, the motor sleeve which houses the propulsion system (rocket motor) was designed in a way to have weight reduction and better heat dissipation. The reason for considering gimbal-based nozzle concept is due to its good reliability, proven performance and the compatibility with additive manufacturing. The design might avoid overhangs beyond 45 degrees (to minimize support structures), incorporate self-supporting angles, or use draft angles for easier removal from the build plate. Features like the fillets being indicate consideration of the printer's resolution and material flow.

Support Images from QuantM:



Weight constraints:

AM allows for complex geometries that reduce material usage, such as lattice structures or hollowed-out sections, which align with the goal of weight reduction. The design might incorporate thin walls or optimized topologies that maintain strength while minimizing mass.

Cost Analysis:

By reducing solid parts, the design likely minimizes material usage. Additionally, consolidating multiple components into a single printed part could have eliminated assembly steps, further lowering costs. The uniform 1 mm fillets suggest a design optimized for printability reducing stress concentrations and are easier to print without support or excessive post-processing.

Post-processing challenges:

Potential issues could include warping or shrinkage of holes. Surface finishing might require post-processing (e.g., reaming, grinding) to meet aesthetic goals. If support were needed for the gimbal’s complex features, their removal could leave minor blemishes.

Estimated cost involved:

Component	Approximate cost
Material (Ti-6Al-4v)	£653

3-D printing	£1,823
Post-processing	£462
Post-processing	£92

Material Considerations and The Quality:

We have researched various materials like steel, PVC, aluminum, titanium to determine which is more convenient to print as 3D model depends on the properties, safety factors etc[7]. Titanium was considered the best material for this project as the weight must be as low as possible. By considering titanium both the properties are satisfied as it has high strength and low weight. Also, titanium should be highly resistant to heat, and its resistance to corrosion is good so overall all these characteristics make it suitable for using this metal to print the model [8].

Effects on design analysis and safety factors:

Safety factors:

Load	FOS	Recommended FOS
Static load	1.4 - 2.2	2.3 - 2.6
Fatigue loading	2.9 - 4.1	4.2 - 5.3
Thermal stress	1.6 - 2.3	2.9 - 3.9

To make the reliability of material good there are several post processing techniques like Hot isostatic pressing, annealing, and precision machining. By using these techniques, it is possible to overcome problems like fatigue, surface roughness.

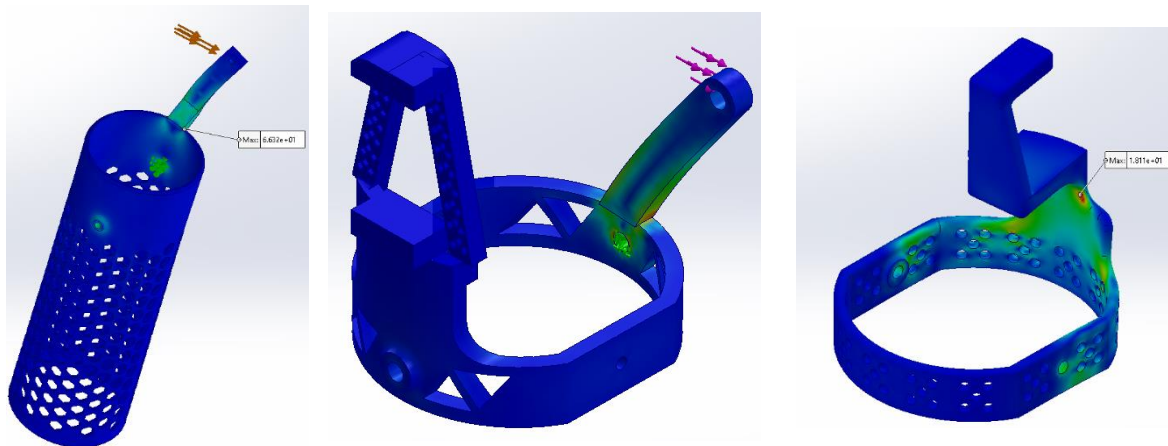
CHAPTER 4:

4.1. Finite Element Analysis (FEA)

After several concept sketches and design iterations, the components were finalized. Using SolidWorks, the parts were simulated on a static study to ensure the strength of the design and analyze the stress distribution. The components were simulated individually with an acting force of 12.1N as this was the maximum possible thrust that the chosen engine, Estes B6-2 can exert during operation [6]. The visual analysis of each individual component simulated can be seen below with the relevant fixtures and forces. All components were assumed to be manufactured in Ti-6Al-4V. A curvature-based mesh was applied to better capture geometry transitions around fillets, slots and structural ribs with all the other mesh parameters left to default settings.

The force vector was applied to the location where the motor would exert thrust against the supporting component as it attempts to correct the trajectory. The components were constrained at the interface point where they are fixed at the pivot mounts where they connect to each other. After the material was applied, the mesh was generated, and six different simulations were performed. One for each component comparing the values between the preliminary design and the finalized one.

Preliminary design



Components Parameters	Outer gimbal	Inner gimbal	Motor sleeve
Max stress (MPa)	2.387e+01	2.000e+01	6.632e+01
Max displacement (mm)	6.647e-02	3.056e-02	6.551e-02
Min safety factor	35	41	12

Final design



Components Parameters	Outer gimbal	Inner gimbal	Motor sleeve
Max stress (MPa)	1.811e+01	1.057e+02	2.662e+02
Max displacement (mm)	4.874e-02	4.113e-02	2.242e-01
Min safety factor	7.2	7.8	3.1

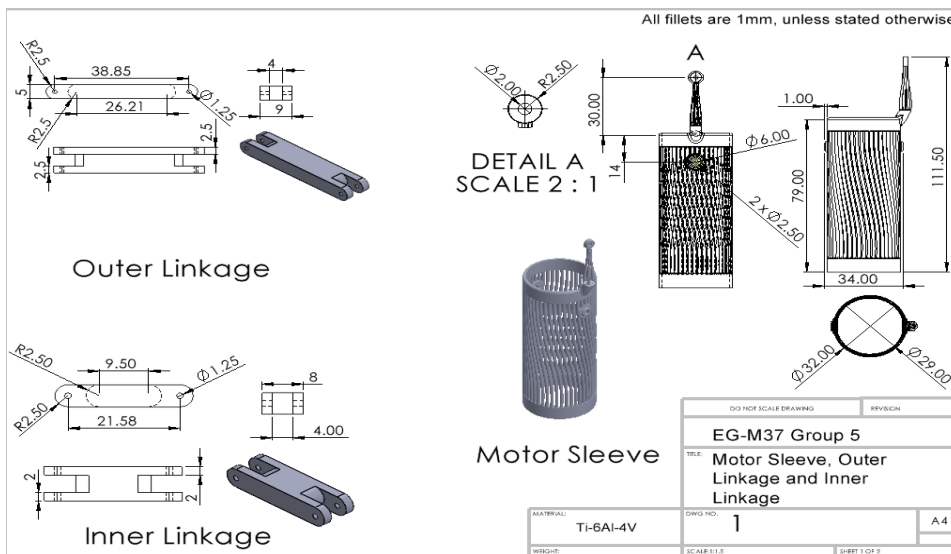
Contribution: My role include i had done a research about the materials which is going to use to build the material and analyze its effects by considering the factor of safety and all . Also preliminary design review

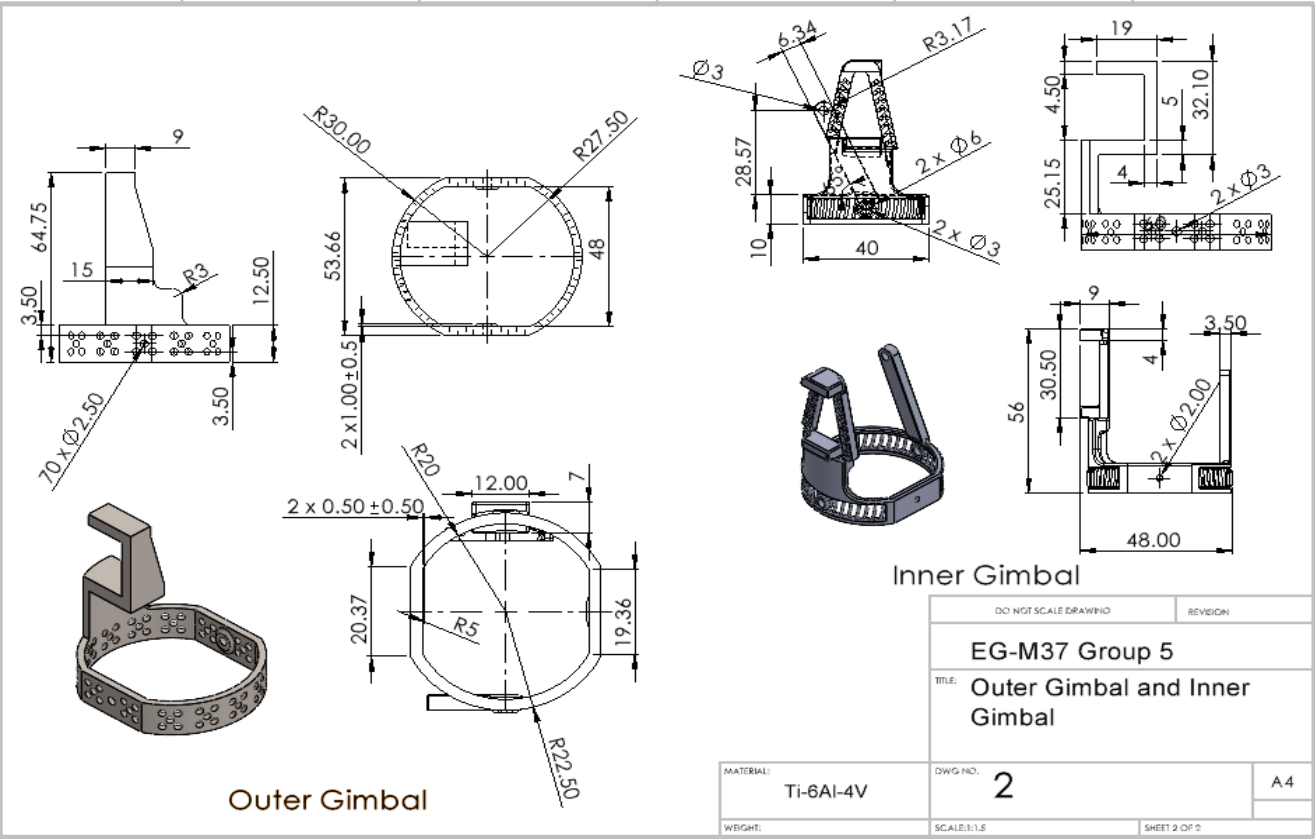
which was my part which is how we are going to consider the design and its aspects , how can we improve its geomatery by various post processing techniques.

4.2. Conclusions and future work

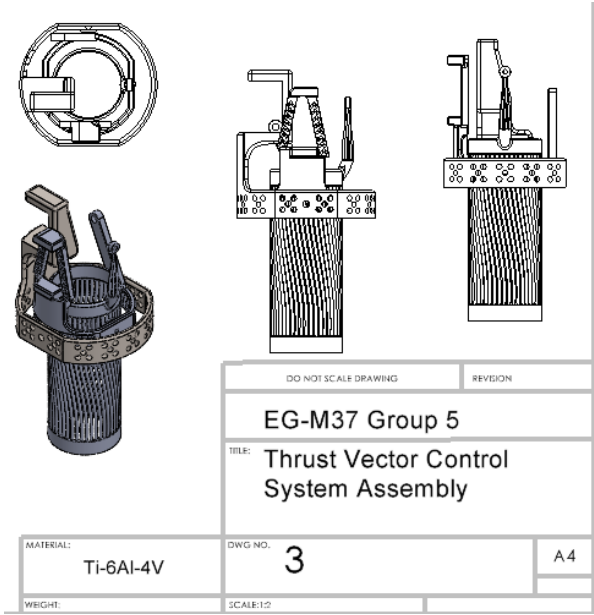
This project successfully demonstrated the design, simulation and manufacturability of compact Thrust Vector Control (TVS). By utilizing the capabilities offered by additive manufacturing, and the material properties of Ti-6Al-4V, the group has managed to produce a lightweight, heat-resistant system that would be challenging to produce with conventional manufacturing methods. Complex geometries were used wherever necessary to enhance the performance while minimizing post-processing requirements ensuring that the components would eventually meet the functional and dimensional requirements for assembly and operation. Furthermore, the principles of additive manufacturing were successfully incorporated throughout the development cycle aiding to overall weight reduction through spiral and hollow features, fillets and lattice structures. Looking forward, future work could consist of an additional dynamic simulation of the entire system-full assembly- during flight conditions, thermal analysis as well as physical testing of the printed system in real life scenarios. One additional approach could include the integration of real-time feedback control using sensors and microcontrollers enhancing overall responsiveness and reliability further increasing the possible scalability of our design as a cost-effective aerospace component.

4.3. Appendix 1: Drawings of 5 parts for Thrust Vector Control System





Appendix 2: Assembly of Thrust Vector Control System



References:

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